

# Operations on Functions

Algebra Worksheet · Grade 9–11

Name: \_\_\_\_\_

Date: \_\_\_\_\_

## Learning Objectives

- Add and subtract functions by combining like terms
- Multiply functions and expand polynomial products
- Divide functions and simplify rational expressions

## Problems

1. Given  $f(x) = 2x + 5$  and  $g(x) = x + 3$ , find  $(f + g)(x)$ .

$$f(x) = 2x + 5, \quad g(x) = x + 3$$

2. Given  $f(x) = 3x + 8$  and  $g(x) = 2x - 12$ , find  $(f + g)(x)$ .

$$f(x) = 3x + 8, \quad g(x) = 2x - 12$$

3. Given  $f(x) = 3x + 8$  and  $g(x) = 2x - 12$ , find  $(g - f)(x)$ .

$$f(x) = 3x + 8, \quad g(x) = 2x - 12$$

4. Given  $f(x) = 4x - 3$  and  $g(x) = x + 7$ , find  $(f - g)(x)$ .

$$f(x) = 4x - 3, \quad g(x) = x + 7$$

5. Given  $f(x) = x + 1$  and  $g(x) = x - 8$ , find  $(f \text{ times } g)(x)$  by expanding the product.

$$f(x) = x + 1, \quad g(x) = x - 8$$

6. Given  $f(x) = x + 1$  and  $g(x) = x^2 - 8x + 7$ , find  $(f \text{ divided by } g)(x)$  and simplify. State any restrictions on  $x$ .





$$f(x) = x + 1, \quad g(x) = x^2 - 8x + 7$$

7. Given  $f(x) = x$  squared minus 1 and  $g(x) = x + 1$ , find  $(f \text{ divided by } g)(x)$  and simplify. State any restrictions on  $x$ .

$$f(x) = x^2 - 1, \quad g(x) = x + 1$$

8. Given  $f(x) = x$  squared plus  $3x + 2$  and  $g(x) = x + 2$ , find  $(f \text{ divided by } g)(x)$  and simplify. State any restrictions on  $x$ .

$$f(x) = x^2 + 3x + 2, \quad g(x) = x + 2$$

9. Given  $f(x) = x$  squared minus  $x$  minus 2 and  $g(x) = x + 1$ , find  $(f \text{ times } g)(x)$  by expanding the product fully.

$$f(x) = x^2 - x - 2, \quad g(x) = x + 1$$

10. Given  $f(x) = x$  cubed plus  $x$  squared minus  $8x$  minus 8 and  $g(x) = x$  squared minus 4, find  $(f \text{ divided by } g)(x)$ , simplify fully, and state all restrictions on  $x$ .

$$f(x) = x^3 + x^2 - 8x - 8, \quad g(x) = x^2 - 4$$



# Operations on Functions — Answer Key

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## Answer Key

### 1. Answer: $(f + g)(x) = 3x + 8$

- Write  $(f + g)(x) = f(x) + g(x) = (2x + 5) + (x + 3)$
- Combine like terms:  $2x + x = 3x$  and  $5 + 3 = 8$
- $(f + g)(x) = 3x + 8$

### 2. Answer: $(f + g)(x) = 5x - 4$

- Write  $(f + g)(x) = (3x + 8) + (2x - 12)$
- Combine like terms:  $3x + 2x = 5x$  and  $8 - 12 = -4$
- $(f + g)(x) = 5x - 4$

### 3. Answer: $(g - f)(x) = -x - 20$

- Write  $(g - f)(x) = g(x) - f(x) = (2x - 12) - (3x + 8)$
- Distribute the negative sign:  $2x - 12 - 3x - 8$
- Combine like terms:  $2x - 3x = -x$  and  $-12 - 8 = -20$
- $(g - f)(x) = -x - 20$

### 4. Answer: $(f - g)(x) = 3x - 10$

- Write  $(f - g)(x) = f(x) - g(x) = (4x - 3) - (x + 7)$
- Distribute the negative sign:  $4x - 3 - x - 7$
- Combine like terms:  $4x - x = 3x$  and  $-3 - 7 = -10$
- $(f - g)(x) = 3x - 10$

### 5. Answer: $(f \cdot g)(x) = x^2 - 7x - 8$

- Write  $(f \cdot g)(x) = (x + 1)(x - 8)$
- Use FOIL:  $x \cdot x + x \cdot (-8) + 1 \cdot x + 1 \cdot (-8)$
- Simplify:  $x^2 - 8x + x - 8$
- $(f \cdot g)(x) = x^2 - 7x - 8$

### 6. Answer: $(f/g)(x) = 1/(x - 7)$ , $x \neq -1$ , $x \neq 7$

- Write  $(f/g)(x) = (x + 1) / (x^2 - 8x + 7)$
- Factor the denominator:  $x^2 - 8x + 7 = (x - 1)(x - 7)$
- Rewrite:  $(x + 1) / [(x - 1)(x - 7)]$
- No common factors cancel; domain restrictions:  $x \neq 1$  and  $x \neq 7$
- Wait — check: denominator factors are  $(x - 1)(x - 7)$ ; numerator is  $(x + 1)$ , no cancellation
- $(f/g)(x) = (x + 1) / [(x - 1)(x - 7)]$ ,  $x \neq 1$ ,  $x \neq 7$

### 7. Answer: $(f/g)(x) = x - 1$ , $x \neq -1$

- Write  $(f/g)(x) = (x^2 - 1) / (x + 1)$
- Factor the numerator using difference of squares:  $x^2 - 1 = (x + 1)(x - 1)$



- Cancel the common factor  $(x + 1)$ :  $(x + 1)(x - 1) / (x + 1) = x - 1$
- Restriction:  $x \neq -1$  (denominator cannot equal zero)
- $(f/g)(x) = x - 1, x \neq -1$

**8. Answer:  $(f/g)(x) = x + 1, x \neq -2$**

- Write  $(f/g)(x) = (x^2 + 3x + 2) / (x + 2)$
- Factor the numerator:  $x^2 + 3x + 2 = (x + 2)(x + 1)$
- Cancel the common factor  $(x + 2)$ :  $(x + 2)(x + 1) / (x + 2) = x + 1$
- Restriction:  $x \neq -2$
- $(f/g)(x) = x + 1, x \neq -2$

**9. Answer:  $(f \cdot g)(x) = x^3 - 3x - 2$**

- Write  $(f \cdot g)(x) = (x^2 - x - 2)(x + 1)$
- Distribute:  $x^2 \cdot x + x^2 \cdot 1 + (-x) \cdot x + (-x) \cdot 1 + (-2) \cdot x + (-2) \cdot 1$
- Expand:  $x^3 + x^2 - x^2 - x - 2x - 2$
- Combine like terms:  $x^3 + (x^2 - x^2) + (-x - 2x) - 2$
- $(f \cdot g)(x) = x^3 - 3x - 2$

**10. Answer:  $(f/g)(x) = (x + 1)/(x - 2) \cdot (x+2)/(x+2)$  simplifies to  $(x^2 + x) / (x - 2)$  — see steps; final:  $(x+1)(x+2)/[(x-2)(x+2)] = (x+1)/(x-2), x \neq \pm 2$**

- Factor the numerator by grouping:  $x^3 + x^2 - 8x - 8 = x^2(x + 1) - 8(x + 1) = (x^2 - 8)(x + 1)$ ... try again: group as  $(x^3 + x^2) + (-8x - 8) = x^2(x + 1) - 8(x + 1) = (x + 1)(x^2 - 8)$
- Factor the denominator:  $x^2 - 4 = (x + 2)(x - 2)$
- Write  $(f/g)(x) = (x + 1)(x^2 - 8) / [(x + 2)(x - 2)]$
- Check for common factors:  $x^2 - 8$  does not factor to include  $(x+2)$  or  $(x-2)$ , so no cancellation
- Restrictions:  $x \neq 2$  and  $x \neq -2$
- $(f/g)(x) = [(x + 1)(x^2 - 8)] / [(x + 2)(x - 2)], x \neq \pm 2$

